



CI4305 COURSEWORK 2A

Design and Build a Clickable Prototype

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TEAM INFORMATION

Module ID	CI4305
Submission	Coursework 2
Team Name	B_TB2_G5

TEAM MEMBERS

KU Number	Name	Contributions
K2432045	Marco Pereira da Silva	Equal
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Accessible link to prototype
Account Creation https://www.figma.com/proto/VmRfqsBrIQUegtFY0W9XgJ/Games-Portal-Backup?node-id=1-18&t=5gwmsYqYHDPqgauS-1
Profile Picture Change https://www.figma.com/proto/VmRfqsBrIQUegtFY0W9XgJ/Games-Portal-Backup?node-id=1-75&t=5gwmsYqYHDPqgauS-1
Purchase Game https://www.figma.com/proto/VmRfqsBrIQUegtFY0W9XgJ/Games-Portal-Backup?node-id=16-1396&t=5gwmsYqYHDPqgauS-1
Play Game https://www.figma.com/proto/VmRfqsBrIQUegtFY0W9XgJ/Games-Portal-Backup?node-id=18-1762&t=5gwmsYqYHDPqgauS-1

CONTRIBUTION NOTES:

If the team members' contributions to the coursework **ARE** equal:

- Enter the word "equal" against each team member's name in the *Contribution* column above.

1 COURSEWORK TEMPLATES

Coursework Templates must be utilised in the design, description and critical analysis of the User Interaction/ User experience design and testing of the prototype.

Week 2 Visual Design Lecture:

Template 1: Visual Design Overview

- Fonts
- Layout
- Colours
- Animation and Video
- Basic Principles for Good Design
- Interface widgets
- Design Trends
- Gestures

Template 2: Basic Principles for Good Design

1. Visibility
2. Affordance
3. Good mapping
4. Feedback
5. Constraints
6. Contrast
7. Consistency
8. Alignment
9. Proximity and grouping

Week 3: Interaction design:

Template 3: Holistic User Experiences

1. Sensual (how absorbed someone is with technology (a game, mobile etc.)
2. Emotional (sorrow, joy, anger, happiness) - angry because something fails, want a phone model, but they are so expensive (emotive)
3. Compositional (how a journey fits together – e.g. online shopping can be a frustrating process), and
4. Spatio-temporal (time – moving or standing still, space – public or private)

Template 4: Usability Goals:

1. “effective to use (effectiveness)
 2. Efficient to use (efficiency)
 3. Safe to use (safety)
 4. Having good utility (utility)
 5. Easy to learn (learnability)
 6. Easy to remember how to use (memorability)”
-

Template 5: User Experience Goals, Desirable and Undesirable aspects of User Design.

Desirable

Satisfying	Helpful	Fun
Enjoyable	Motivating	Provocative
Engaging	Challenging	Surprising
Pleasurable	Enhance Sociability	Rewarding
Exciting	Supporting Creativity	Emotionality Fulfilling
Entertaining	Cognitively stimulating	

Undesirable

Boring	Unpleasant
Frustrating	Patronizing
Making One feel guilty	Making one feel stupid
Annoying	Cutesy
Childish	Gimmicky
Entertaining	

Week 4 Information Architecture:

Template 6: Metadata

Template 7: LATCH

Template 8: Information Seeking

Week 5 Cognitive Aspects:

Template 9: Cognitive processes

1. attention
 - 2 perception
 - 3 memory
 - 4 learning
 - 5 reading, speaking and listening
 - 6 problem-solving, planning, reasoning, and decision-making.
-

Week7 testing and evaluation.

Template 10: Usability Problems

Might annoy or displease users.

(SATISFACTION)

- Might prevent users from finishing tasks in a reasonable timeframe

(EFFECTIVENESS & EFFICIENCY)

- Might make users misunderstand something.

(EFFECTIVENESS & EFFICIENCY)

- Might be difficult to understand, figure out, or learn.

(EFFECTIVENESS & EFFICIENCY)

Words, phrases, or names (on the interface) that:

- Users don't understand.
- Users misunderstand.
- Have different meanings to different people.
- Users just don't like or will not use.

Tasks that users may want to do:

- But can't do.
- But take a long time to finish.
- But can't achieve without help or hints
- But are not intuitively logical for them
- e.g., wrong order of doing things.

Template 11: cognitive walkthrough process questions:

1. Does the user understand that a specific action is required to achieve their goal?
2. Is the action visible and will the user notice that the correct action is available?
3. Will the user recognise the action as the correct one?
4. Does the user get appropriate feedback, and will they understand it?

Template 12: Heuristic Evaluation (HE)

List of Heuristics:

- H1. Visibility of system status
- H2. Match between system and the real world
- H3. User control and freedom
- H4. Consistency and standards
- H5. Error prevention
- H6. Recognition rather than recall
- H7. Flexibility and efficiency of use
- H8. Aesthetic and minimalist design
- H9. Help users recognise, diagnose and recover from errors.
- H10. Help and documentation

Also To help you further you can [Check out Jakob Nielsen's 10 Usability Heuristics for User Interface Design and download the poster](#)

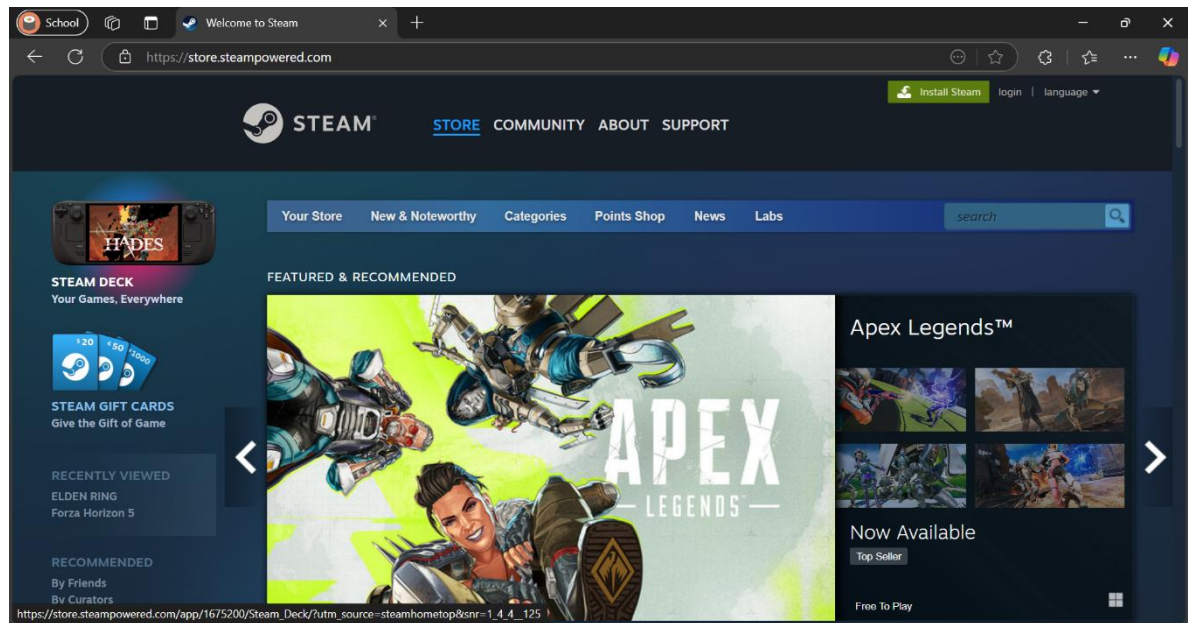
2.1 – 2.5 VISUAL DESIGN

2.1 INSPIRATION SCREENSHOTS

Complete this section in accordance with the coursework marking scheme and workshop tasks.

This section will include screenshots from sites, apps, images or other sources that serve as an inspiration for an aspect of your app design. Describe why each screenshot has been chosen and what aspect will relate to your app. MAXIMUM 5 screenshots.

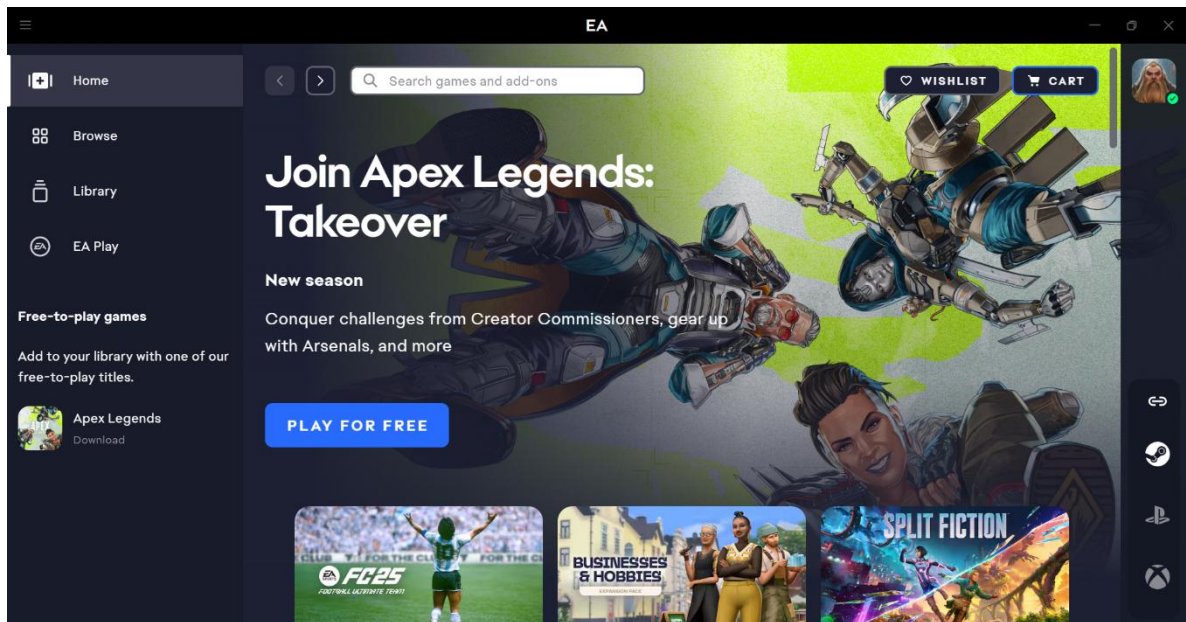
SCREENSHOT 1



Steam - <https://store.steampowered.com/>

We drew inspiration from the Steam app and website due to their clean and minimalistic design, particularly in the header and game display elements. Their interface is concise, intuitive, and accessible to users of all experience levels.

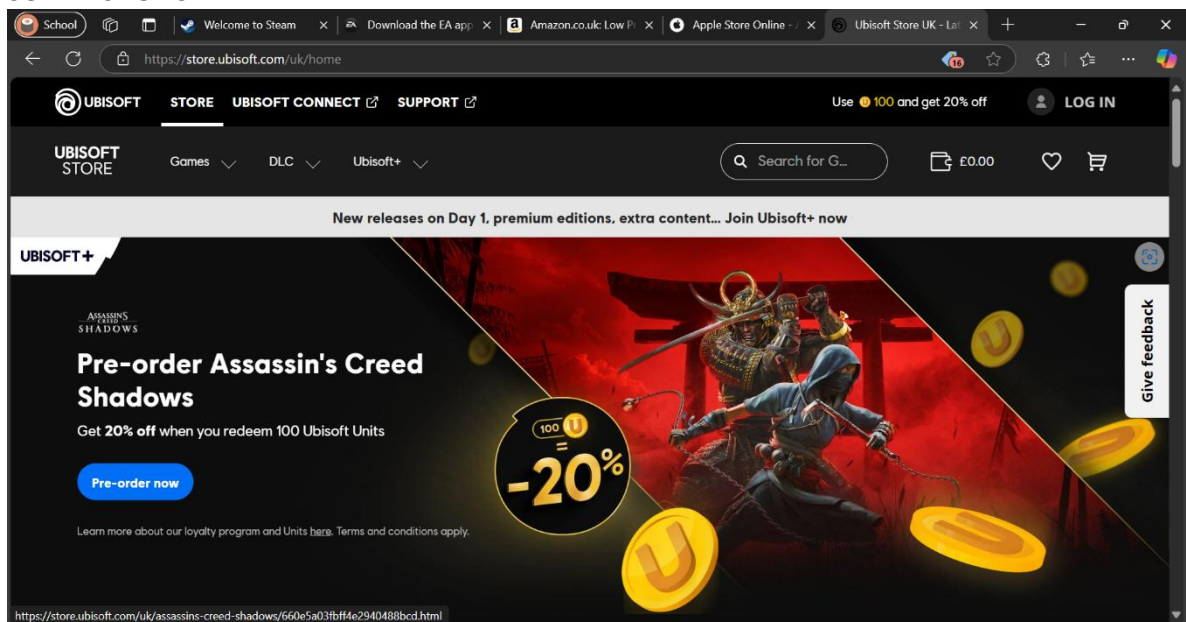
SCREENSHOT 2



EA - <https://www.ea.com/en-gb/ea-app>

The EA Games app influenced our overall layout, particularly its use of floating text and elements. This design choice creates a sense of openness and ease, preventing the interface from feeling cluttered with excessive menus, elements or separators.

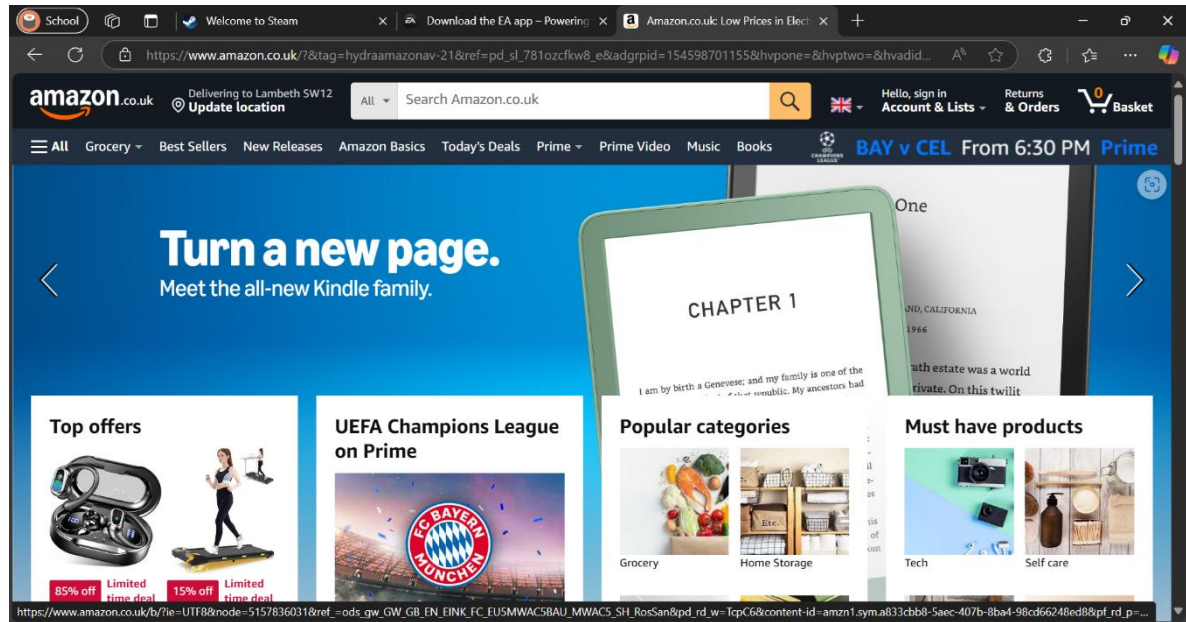
SCREENSHOT 3



Ubisoft - <https://store.ubisoft.com/uk/home>

The Ubisoft website inspired our choice of colours and fonts. It effectively utilises colour contrast throughout the site, while the font size and colour complement the overall design style seamlessly.

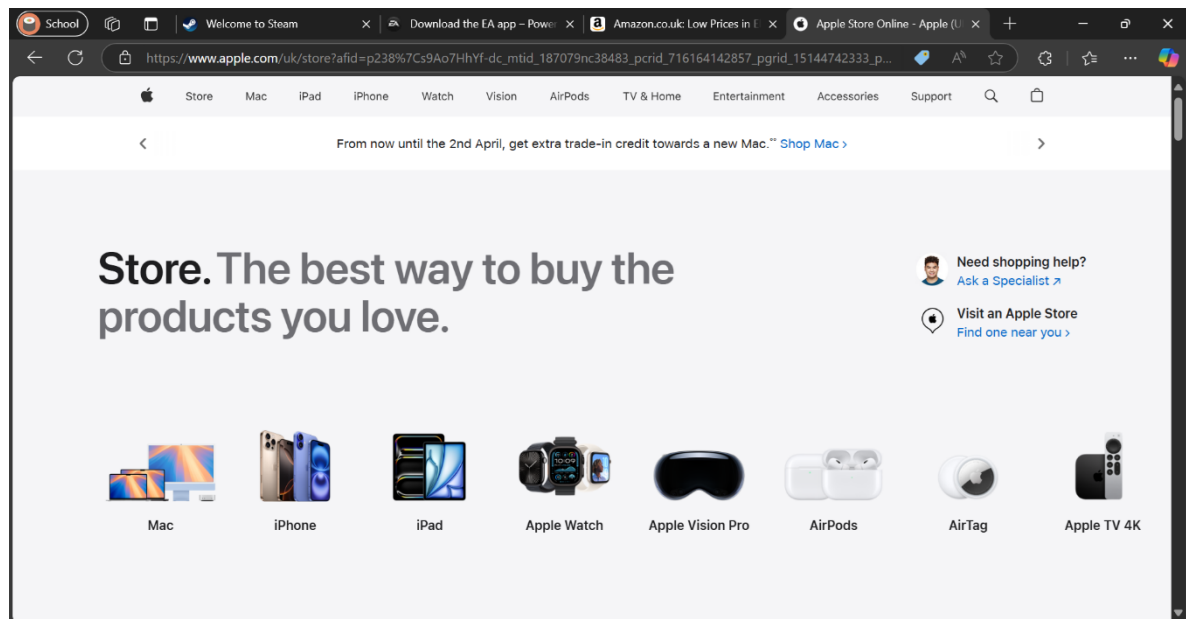
SCREENSHOT 4



Amazon - https://www.amazon.co.uk/ref=nav_logo

The Amazon website influenced our approach to product displays. As one of the largest online retailers, Amazon must present its products efficiently. The use of flipping cards and floating blocks enhances usability, making product browsing intuitive and accessible for all users.

SCREENSHOT 5



Apple Store - <https://www.apple.com/uk/store>

The Apple Store inspired our header menu design, showcasing a large amount of information in a clean and organised manner. Additionally, the use of floating elements throughout the page enhances the user experience by creating a sense of openness without overwhelming the visual layout.

2.2 COLOUR PALETTE SCREENSHOTS

Complete this section in accordance with the coursework marking scheme and workshop tasks. This section will include colour palette screen shots. These are the colours that will be used to define the colours used in your app, or that are prevalent in your game design (e.g. characters, backgrounds etc).



In our app we used this colour pallet for the following:



#680F42 and #27084B is used in the app background.

We chose these colours because they represent creativity and luxury and convey depth and elegance. Commonly linked to technology and futuristic themes, they give a premium and immersive feeling, minimising eye strain, especially in low-light environments, making them ideal for a gaming focused application.

The subtle reference to the British flag through these muted reds and blues adds a unique branding element, helping the app stand out with a meaningful visual identity.



#FFFFFF and #000000 is used in text.

These colours are ideal for contrasting text from the background allowing for better readability and reducing eye strain.

Pure white on black or vice-versa, can cause visual fatigue due to high contrast, so slight variations are used for a softer, more comfortable reading experience.



#9E1563 and #FEB97E used for buttons and highlights.

These colours bring vibrancy and warmth to the design. The deep pink adds a bold and energetic touch when pressing a button, while the peach tone introduces a softer, more welcoming contrast. They offer a modern and engaging aesthetic, adding emphasis without overwhelming the user interface.



#000000 with 50% transparency for the menus and block elements.

This creates a sleek and modern design while maintaining focus on content. The semi-transparent black provides depth without being too overpowering, allowing the background and underlying elements to subtly show through.



#EE0004 is used to highlight important user actions.

The bright red colour is an effective choice for highlighting important actions, such as deleting account. Red is universally associated with urgency and danger, making it ideal for actions that require careful consideration.



#6A47C7 used in the rim around the profile picture.

This colour adds a distinctive and modern touch, enhancing the overall aesthetic without being overly distracting.

2.3 LOGOS, ICONS, CHARACTERS, PROPS, BACKGROUNDS

Complete this section in accordance with the coursework marking scheme and your workshop tasks. This section will include draft logos, icons, characters, props, backgrounds etc., that will be used as part of your app design.



This logo represents the core identity of our app, “Games Portal”. Designed with the app’s name in mind, the controller symbolises the gaming platform, while the circular gradient background suggests a portal. The controller appears to be crossing through the portal, symbolising the transition to an futuristic and immersive gaming experience.



This icon is featured in the language selection dropdown menu, clearly indicating to the user that it is related to language settings. Its intuitive design ensures users can easily recognise its function.



This is the iconic trash bin icon, used to indicate that pressing the button will delete the game from the shopping cart. Its familiar design provides a clear and intuitive action for users.



This is a stock profile picture that the user can change into in the application.



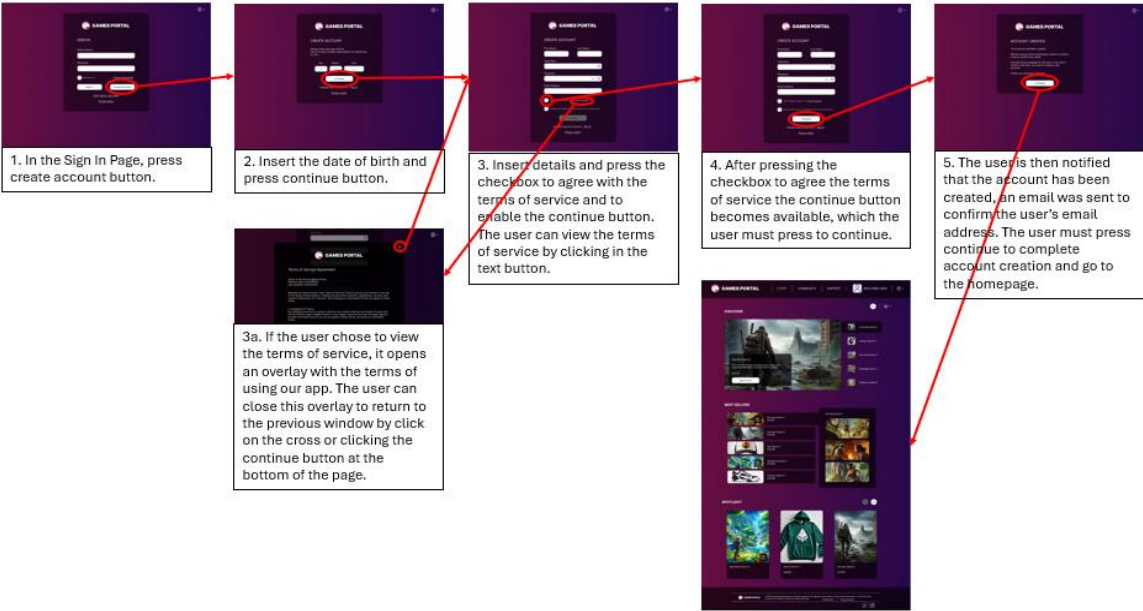
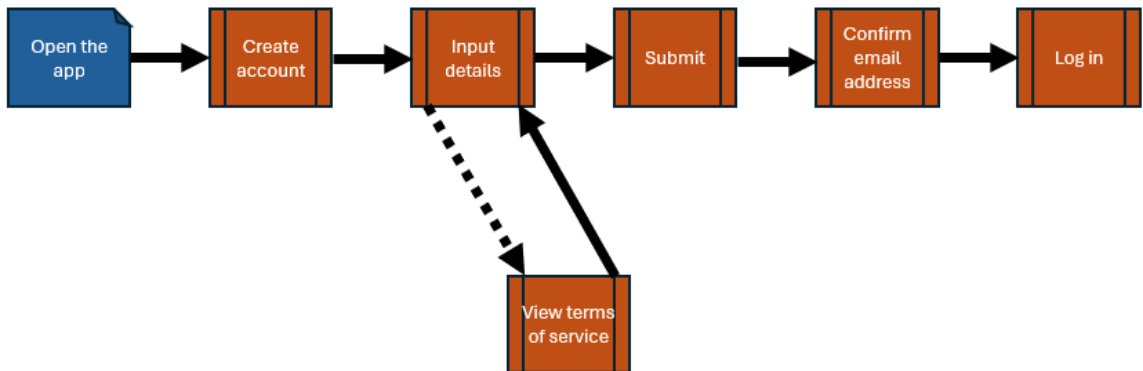
This is an AI-generated game image used in our platform. This image relates to a survival game.

2.4 – 2.5 MEDIUM FIDELITY PROTOTYPE

Complete this section in accordance with the coursework marking scheme and your workshop tasks.

Accessible link to the prototype:

A presentation of 4 clickable walkthroughs that enable a user to achieve 4 different goals.

Aim/Goal of walkthrough 1
The user creating a new account on the app.
Relevant functional requirements
Register user, view policies, verify email, access data, store data, login user
Provide a set of annotated screenshots that illustrate this walkthrough
 1. In the Sign In Page, press create account button. 2. Insert the date of birth and press continue button. 3. Insert details and press the checkbox to agree with the terms of service and to enable the continue button. The user can view the terms of service by clicking in the text button. 3a. If the user chose to view the terms of service, it opens an overlay with the terms of using our app. The user can close this overlay to return to the previous window by click on the cross or clicking the continue button at the bottom of the page. 4. After pressing the checkbox to agree the terms of service the continue button becomes available, which the user must press to continue. 5. The user is then notified that the account has been created, an email was sent to confirm the user's email address. The user must press continue to complete account creation and go to the homepage.
Supporting evidence:
 <pre>graph LR; A[Open the app] --> B[Create account]; B --> C[Input details]; C --> D[Submit]; D --> E[Confirm email address]; E --> F[Log in]; C -.-> G[View terms of service]; G --> C;</pre>

Aim/Goal of walkthrough 2

The user changing the profile picture on the account.

Relevant functional requirements

View profile, modify profile, modify profile picture, access data, store data

Provide a set of annotated screenshots that illustrate this walkthrough

1. Go to the navigation menu and press welcome user to access the profile options.

2. Press on my profile.

3. Click on change profile button to change profile details.

4. This opens an overlay where the user must click on change picture to open dropdown menu.

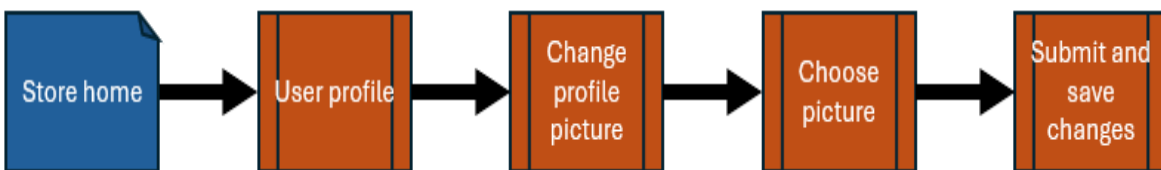
5. Click on stock image to open the stock profile image selector overlay.

6. Click on the profile picture you want to change to.

7. Click on accept button to accept the chosen profile picture.

8. The user must click on save changes button to permanently change the profile picture on their profile.

Supporting evidence:



Aim/Goal of walkthrough 3

The user buying a game and this game to be added to the user library for future use.

Relevant functional requirements

View game, add to cart, buy game, access data, store data, verify card, charge card

Provide a set of annotated screenshots that illustrate this walkthrough

1. Click on the game you want to view.

2. Click on add to cart to add game to the cart or, click on buy game to jump to payment.

3. If the game was added to the cart, a "1" notification will show near the cart icon. Click on the cart icon to open the cart or click buy now to jump to payment.

4. If the cart icon was pressed this opens the cart dropdown menu with all the information for purchase. Click checkout to go to payment.

5. This opens an overlay where the user must select a payment type. Only then the pay now button will become available.

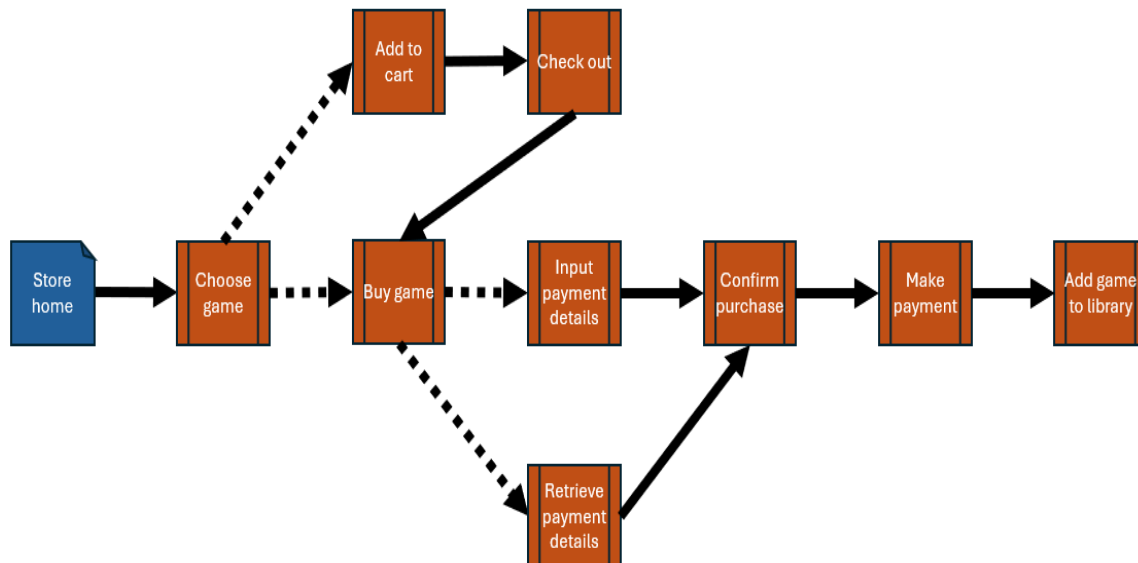
6. After selecting a payment type and inserting payment details, click on confirm button to confirm payment and enable pay now button.

7. The pay now button becomes available which the user must click to proceed with the payment.

8. The payment is being confirmed by the system and once it finishes, moves on to the next step.

9. The payment is confirmed and done successfully. The user is informed that the game is added to their library and an email is sent to the user with a receipt for the transaction.

Supporting evidence:



Aim/Goal of walkthrough 4

The user opening the user library, downloading, installing and playing a game.

Relevant functional requirements

View library, download game, install game, access data, store data, play game

Provide a set of annotated screenshots that illustrate this walkthrough

1. Go to the navigation menu and press welcome user to access the profile options.

2. Press on library to open the user library.

3. On the corner of each game image there is a small information about if the game is on the computer already or if needs to download. Click on that icon to download/play the game or click on the game you want to view to open its page.

4. Click on download button to download and install the game on the computer.

5. The user can choose the destination to save the game files and must press continue to start downloading and installing the game.

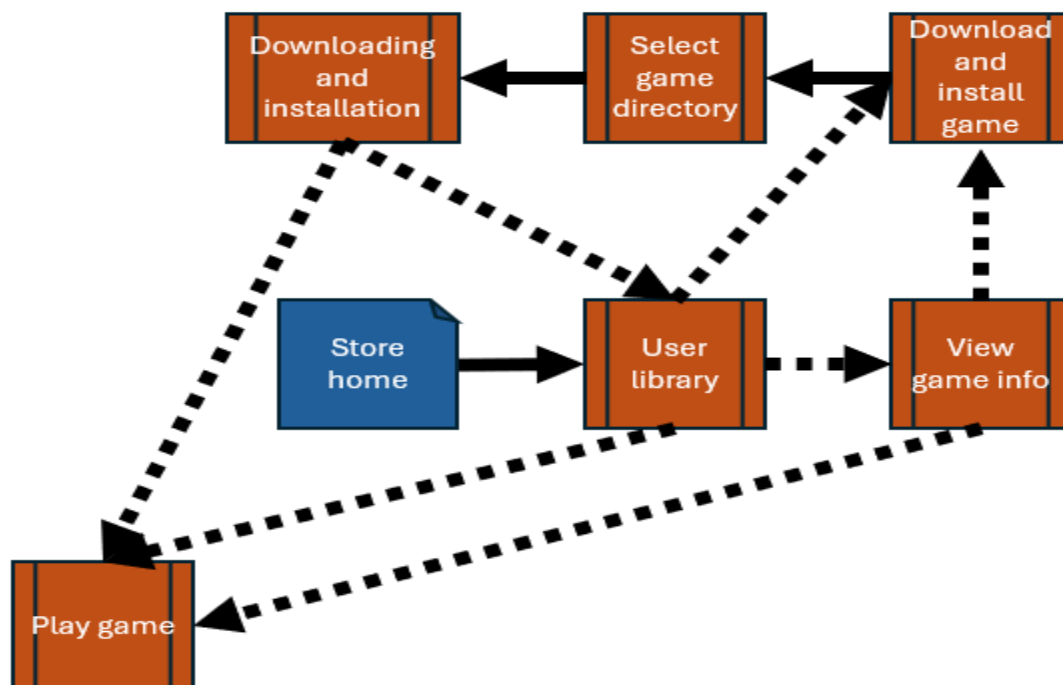
6. The game is being downloaded and installed. The user can cancel this at any time.

7. After the game is installed, the user can choose to play the game or return to the library.

8. If return to library was chosen, it returns to the library and updates information icon. Click icon to play the game or click game image to open the game page.

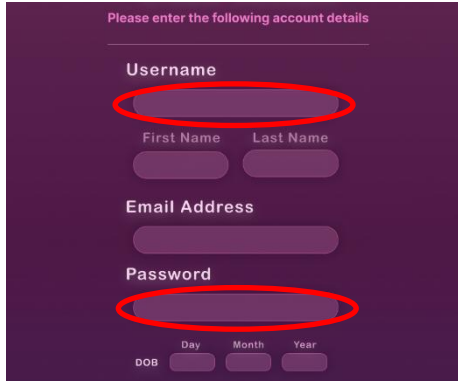
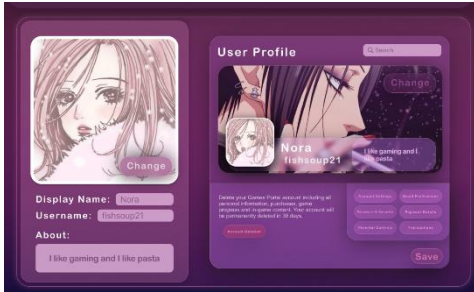
9. If game image was pressed it opens the game page. Click play game button to open the game.

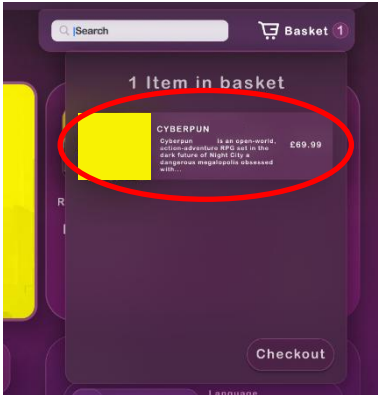
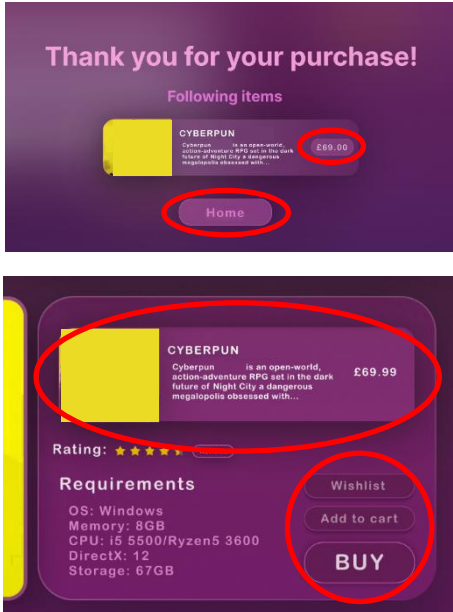
Supporting evidence:



2.6 – 2.8 HEURISTIC EVALUATION

Complete this section in accordance with the coursework marking scheme.

Problem ID	Rich Description, referenced action in prototype, related documents	Heuristic ID	Severity Rating
P1	This issue occurs during the account creation process when users are prompted to input details, like username and password. The input fields cannot be left blank and require specific criteria (usernames must be 6 to 20 characters long, and passwords must include at least one uppercase letter and one number). However, the system does not provide real-time guidance to users about inputs in this fields. This lack of guidance and recognition may lead to confusion and frustration, potentially causing users to abandon the registration process. 	H6	R
P2	This issue occurs on the user profile page, when the user is trying to change the profile picture. Even though the picture has been edited in the profile section it does not change in the top right corner of the page. The layout appears cluttered, and the navigation menu blends into the overall design rather than having a distinct placement. The lack of clear separation makes it difficult for users to locate buttons and other elements, leading to confusion. The mispositions and number of items on the page may annoy users, forcing them to spend extra time searching, which can result in frustration and dissatisfaction. 	H8	Q

P3	This issue occurs during the purchasing process when a user adds a game to the cart. If they accidentally select the wrong game, there is no option to remove it directly from the cart menu. Instead, users must navigate elsewhere in the application to remove the unwanted item, which adds unnecessary steps and frustration. This lack of convenience and freedom may lead to dissatisfaction, discouraging users from completing their purchase or engaging with the platform altogether. 	H3	R
P4	This issue affects the overall app design, particularly when users interact with buttons to complete actions. The inconsistent button design, which closely resembles other elements in the application, makes it difficult for users to distinguish interactive elements from static ones. While not a critical issue, this lack of consistency and visual clarity can cause confusion and reduce user engagement, potentially discouraging them from noticing or completing actions within the app. From a design perspective the colouring of text resembles too much off the background colour, which can make text hard to see. 	H4	P

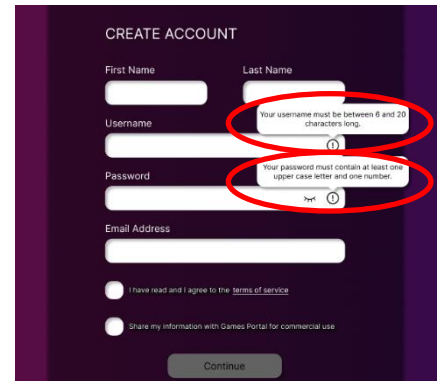
2.9 – 2.10 CRITICAL REVIEW AND DESIGN RECOMMENDATIONS

Complete this section in accordance with the coursework marking scheme.

Redesign recommendation 1 (H6)

Recommendations in bullet points

- Display clear and informative messages near the input fields, specifying the input field criteria.
- Make use of intuitive icons to display the information.
- Avoid generic text, be specific and helpful to avoid confusion.



The screenshot shows a 'CREATE ACCOUNT' form with fields for First Name, Last Name, Username, Password, and Email Address. Below the Username field, a red circle highlights the message: 'Your username must be between 6 and 20 characters long.' Below the Password field, another red circle highlights the message: 'Your password must contain at least one upper case letter and one number.' There are also checkboxes for 'I have read and I agree to the terms of service' and 'Share my information with Games Portal for commercial use', and a 'Continue' button at the bottom.

Justification by reference to evaluation

In reference to Heuristic Evaluation P1, we implemented tooltips for the username and password input fields.

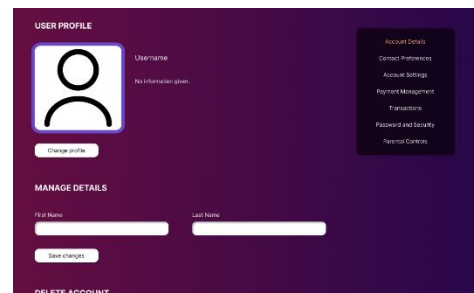
These tooltips provide users with clear criteria for entering valid credentials, eliminating the need for guesswork. They feature a universally recognised information icon, ensuring intuitive interaction. The tooltip content appears when users hover over the icon, offering guidance without cluttering the interface.

This enhancement minimises confusion and improves the overall user experience.

Redesign recommendation 2 (H8)

Recommendations in bullet points

- Ensure navigation menu has a distinct placement, separated from the main content.
- Reduce clutter on the page by grouping related actions into tabs.
- Use whitespaces effectively to create a cleaner layout and improve readability.



The screenshot shows a 'USER PROFILE' page. On the left, there is a profile picture placeholder and a 'Change profile' button. Below this is a 'MANAGE DETAILS' section with input fields for 'First Name' and 'Last Name', and a 'Save changes' button. At the bottom is a 'DELETE ACCOUNT' button. On the right, there is a side navigation menu with links: 'Account Details', 'Current Preferences', 'Account Settings', 'Payment Management', 'Transactions', 'Password and Security', and 'Personal Details'.

Justification by reference to evaluation

In reference to the Heuristic Evaluation P2, we redesigned the user profile page and the overall app layout to enhance readability and user comprehension.

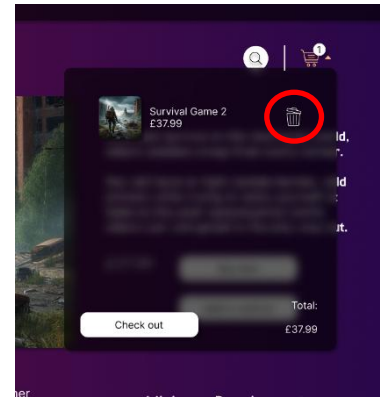
Similar actions are now grouped within a fixed side tab that remains visible while scrolling, ensuring easy access at all times. User input fields have been reorganised into logical sections to minimise confusion and indicate related actions.

Additionally, improved use of whitespace reduces visual clutter, creating a cleaner and more user-friendly experience.

Redesign recommendation 3 (H3)

Recommendations in bullet points

- Add a clear remove button within the cart menu, allowing users to delete unwanted items directly.
- Use a universally recognised bin icon to enhance visibility and user understanding.



Justification by reference to evaluation

In reference to the **Heuristic Evaluation P3**, we redesigned the shopping cart by adding a universally recognised bin icon, allowing users to remove games directly from the cart.

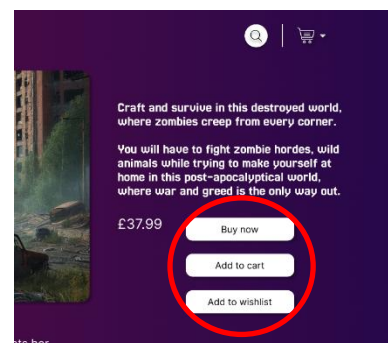
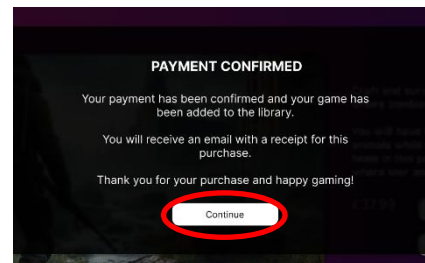
This improvement enhances interactivity and intuitiveness, giving users more control over their purchases without requiring additional navigation.

By simplifying this process, we reduce frustration and improve the satisfaction of the overall shopping experience.

Redesign recommendation 4

Recommendations in bullet points

- Ensure all buttons follow a consistent design, including size, colour and font.
- Differentiate buttons from static elements by using distinct visual cues such as contrasting colours, borders or hover effects.
- Use primary, secondary and tertiary to guide user effectively.
- Use clear and action-oriented text on buttons, such as “Buy now” instead of “Next”.



Justification by reference to evaluation

In reference to the Heuristic Evaluation P4, we standardised all buttons throughout the application by using the same shape, colour, and text dimensions.

This consistency ensures that buttons are easily identifiable and visually distinct from other elements on the screen.

By maintaining a uniform design for all interactive elements, we not only create a clear and structured page but also enhance the user's ability to quickly recognise clickable items, improving navigation efficiency and reducing any misunderstandings regarding the process of purchasing a game start to finish. As a result, users can engage with the app more intuitively, leading to a smoother and more seamless experience.

TEAM CONCERN STATEMENT

This section only needs to be completed if you have concerns regarding:

- Loss of contact with a team member
- Limited contributions to team work
- Non-submission of individual work

Please complete the details below.

KU Number	Lost contact?	Contribution to group work (%) Enter an estimate between 0% and 100%.	Attended meetings?	Received individual course work?
	YES/NO		YES/NO	YES/NO
	YES/NO		YES/NO	YES/NO
	YES/NO		YES/NO	YES/NO
	YES/NO		YES/NO	YES/NO
	YES/NO		YES/NO	YES/NO
	YES/NO		YES/NO	YES/NO

Where subsequent concerns are clarified, the marks for the contributing members of the group will not be affected.